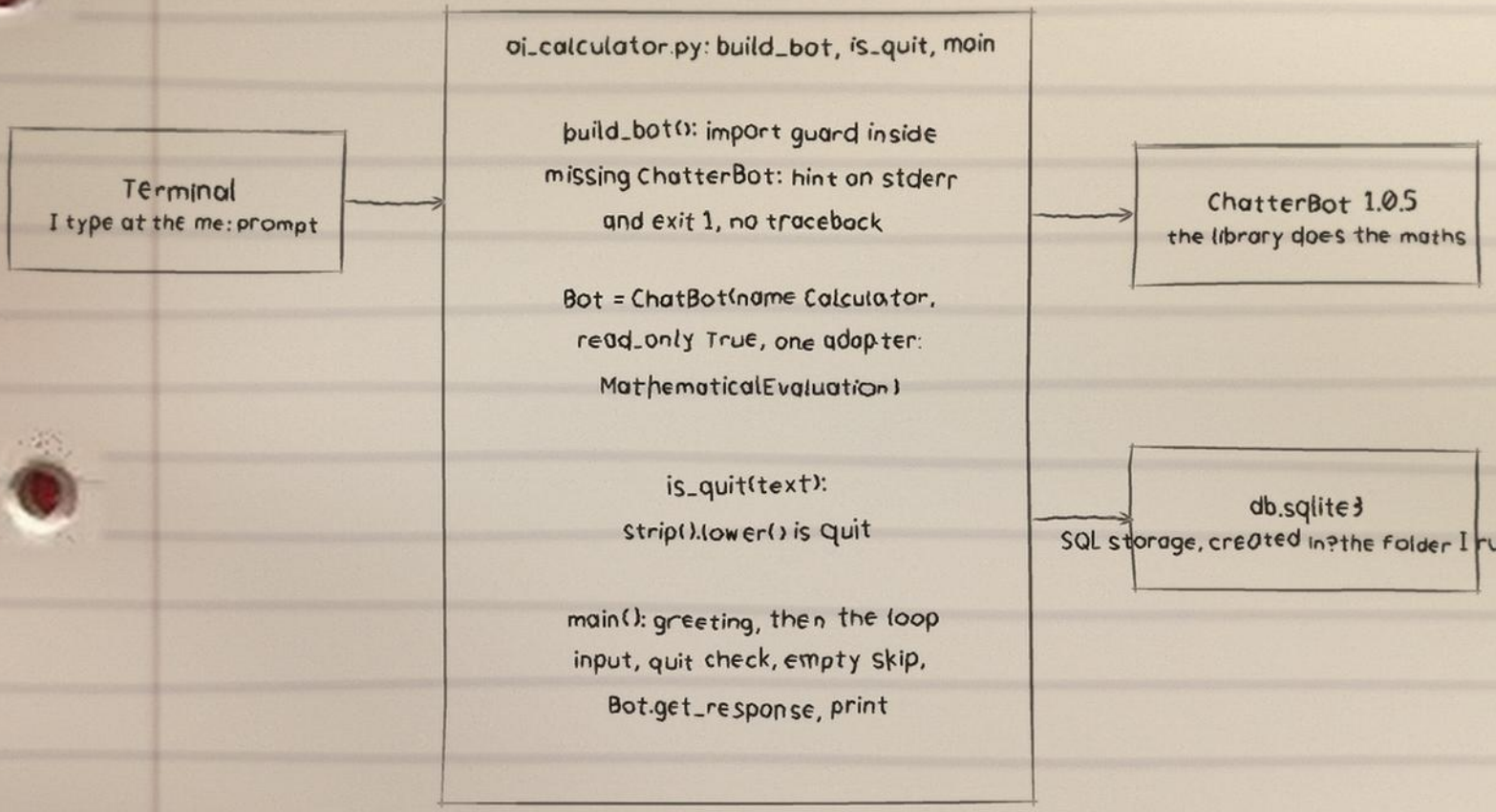
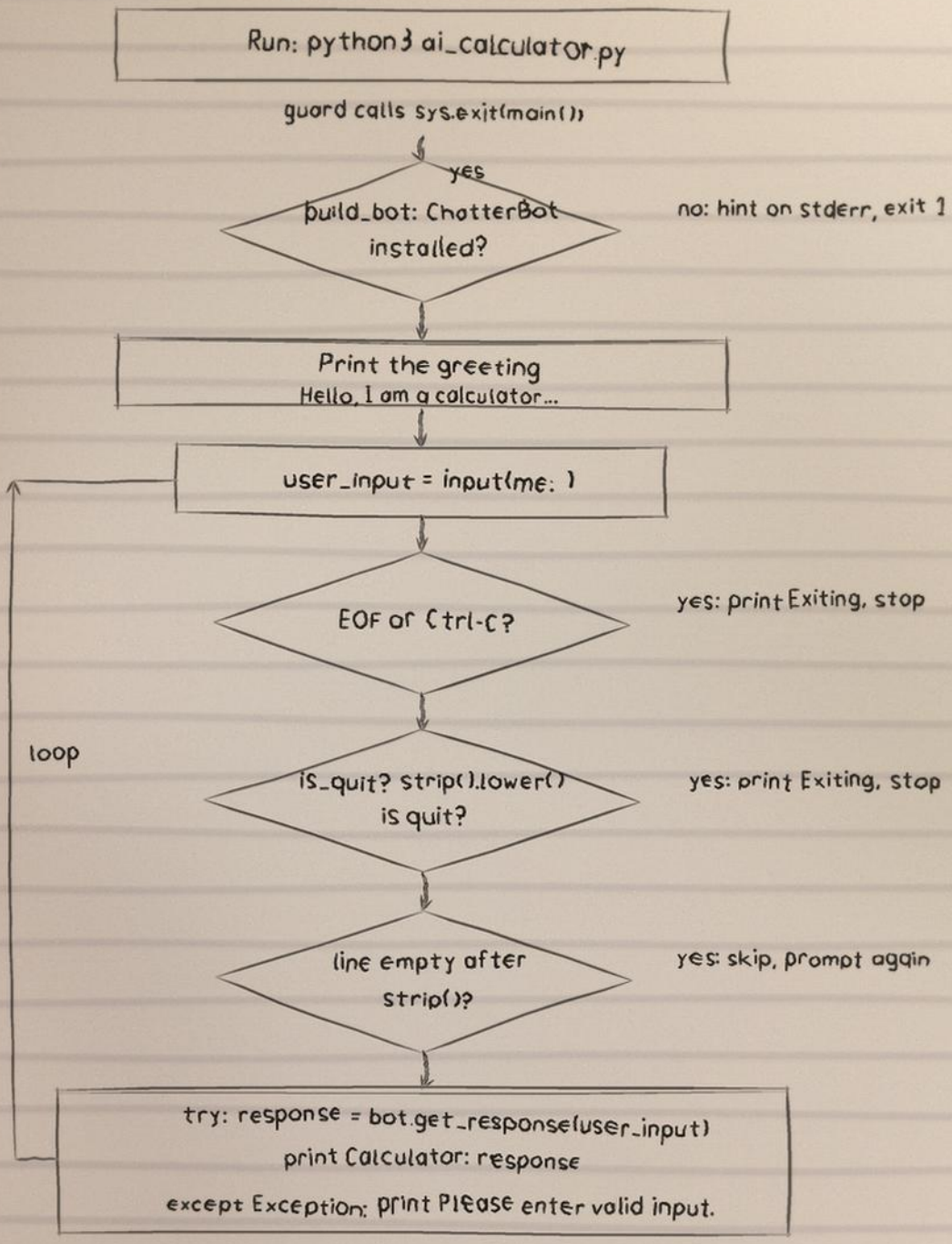
AI-Calculator architecture: three functions, one library



Entry point: if __name__ is __main__, sys.exit(main()). Importing this file now starts nothing, so tests can import it.
read_only matters: the bot does not learn from what gets typed at it.

one script, signed GR

The loop in main(), and its exits



Every exit prints the same word, Exiting, by design. main() returns 0 on a normal end.

The wide except also hides library failures: the README records a run where that was every answer.

Product map: a calculator that chats, maths only

AI-CALCULATOR: type a maths question, get an answer in the terminal

WHO IT SERVES	WHAT IT DOES	WHAT IT DOES NOT DO	WHERE IT STOPS
Me, at a terminal, wanting a quick answer without leaving it.	main() hands each non-empty line to MathematicalEvaluation and prints its reply.	No general chat: no other logic adapter is loaded. It does not learn.	Empty lines are skipped. Bad input gets Please ente valid input, and the loop goes

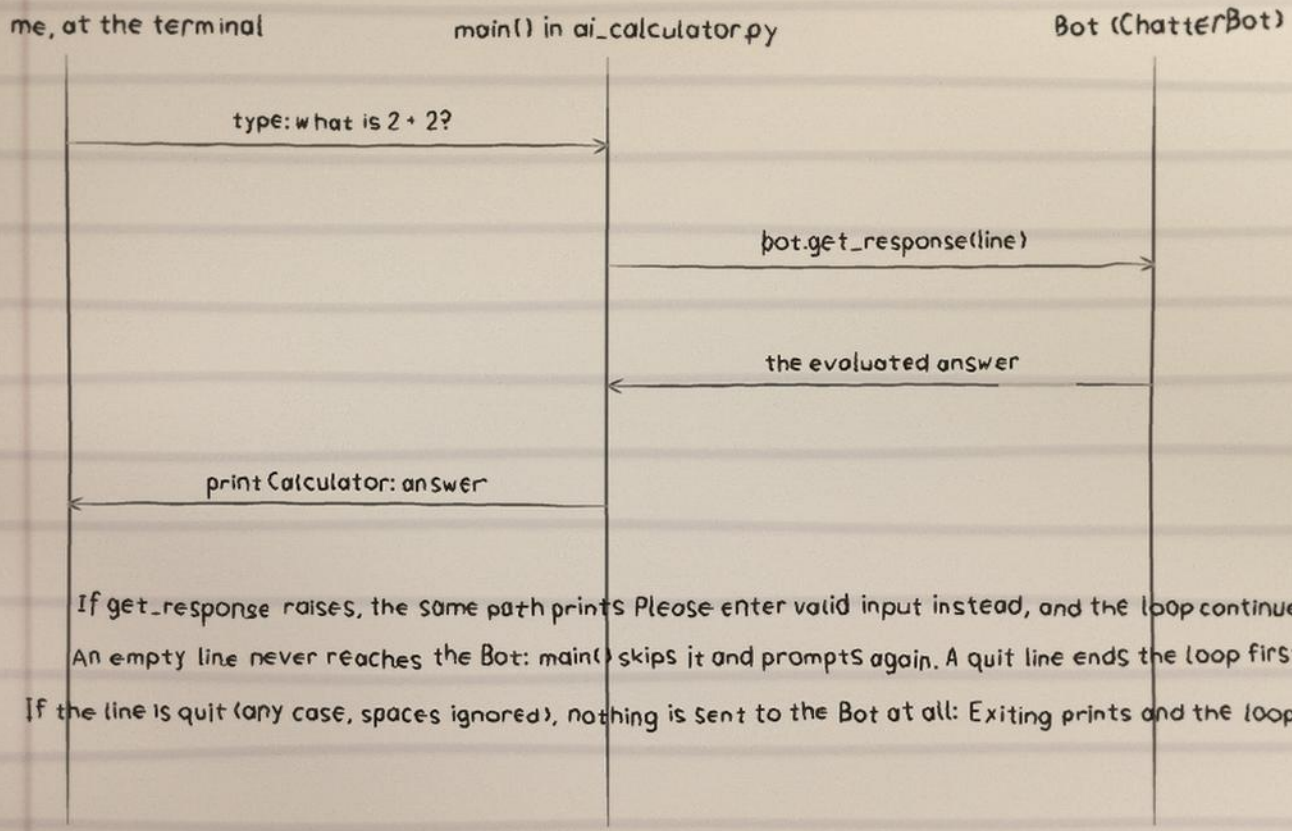
Inputs: lines typed at me: , ending with quit, Ctrl-D or Ctrl-C. Outputs: Calculator: lines on stdout, and a db.sqlite3 file le

Needs Python 3 and ChatterBot (requirements.txt). First use downloads NLTK data, so it needs the network once.

New at this head: importing the file starts nothing, and test_ai_calculator.py covers quit matching and the missing install.

counter card, GK

Sequence: one question, from prompt to answer



If get_response raises, the same path prints Please enter valid input instead, and the loop continues.

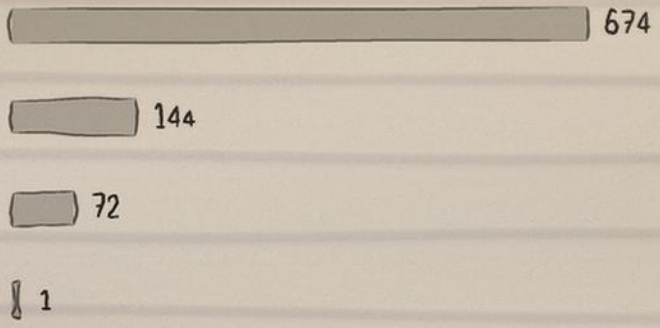
An empty line never reaches the Bot: main() skips it and prompts again. A quit line ends the loop first.

If the line is quit (any case, spaces ignored), nothing is sent to the Bot at all: Exiting prints and the loop ends.

sequence noted, GK

AI-Calculator in numbers, counted 2026-10-30

Lines by file kind over tracked files, images excluded. The LICENSE is most of this repo by line count.



7 tracked files, 6 commits. First commit 2023-02-11, last 2026-10-30, from git log.

Python split: ai_calculator.py 99 lines, test_ai_calculator.py 45 lines. Tests: 4 unittest tests ran here, all passed.

The program file is 99 lines: build_bot, is_quit and main, plus the entry guard.

tallied, GK

AI Calculator: schematic of the apparatus

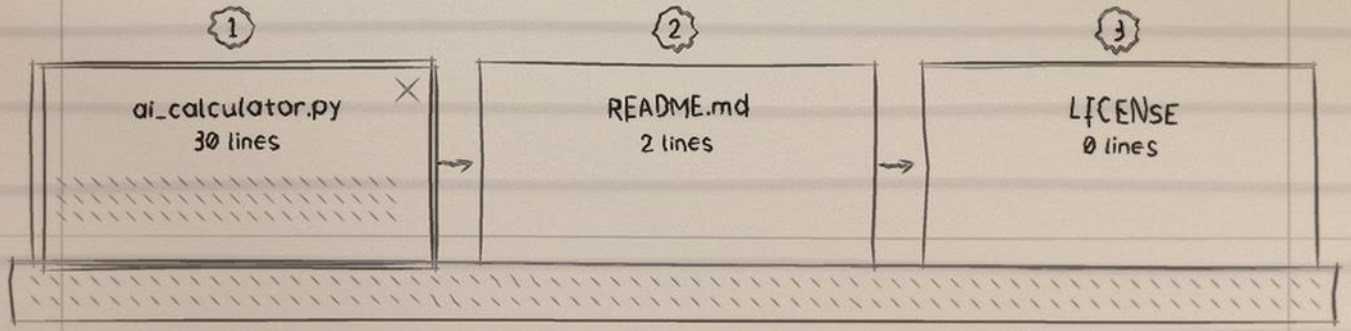
the system as a machine: parts, flow, and measurements

construction lines left in, as drawn

parts list

1. ai_calculator.py
2. README.md
3. LICENSE

in flow order, first to last



base plate: the repo itself, sectioned

span: 3 files, 32 lines counted

turns of the crank so far: 2 commits, 2023-02-11 to 2023-02-11

measured and sketched, GK